Notes on the theoretical framework for generalized new neutrino interactions

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Abstract

These notes outline a model-independent framework for simulating the propagation of high-energy astrophysical neutrinos under the influence of generalized, flavor-biased secret interactions. By decoupling the coherent long-range matter potential (sourced by macroscopic particle distributions) from the incoherent scattering (sourced by the Cosmic Neutrino Background), we establish a unified Quantum Kinetic Equation (QKE). This framework bypasses specific UV-completions in favor of a generalized Hermitian 3×3 EFT coupling matrix, suitable for direct confrontation with IceCube HESE flavor and spectral data.

1 The generalized EFT Lagrangian

To remain agnostic to the underlying gauge symmetry, we parameterize the new physics via effective four-fermion contact operators. We define two distinct interaction sectors: one for non-standard self-interactions (NSSI) with the Cosmic Neutrino Background (CνB), and one for long-range interactions (LRI) with ordinary matter distributions.

The NSSI Lagrangian for active left-handed neutrinos ν_α ($\alpha \in \{e, \mu, \tau\}$) is given by

$$\mathcal{L}_{\rm NSSI} = \sum_{\alpha,\beta,\gamma,\delta} \frac{G_{\alpha\beta\gamma\delta}}{\sqrt{2}} (\overline{\nu}_\alpha \gamma^\mu P_L \nu_\beta)(\overline{\nu}_\gamma \gamma_\mu P_L \nu_\delta). \tag{1}$$

For phenomenological tractability, we compress the four-flavor tensor into an effective 3×3 coupling matrix, $\mathbf{G}_{\nu\nu}$. To preserve unitarity and ensure a real effective Hamiltonian, $\mathbf{G}_{\nu\nu}$ must be explicitly Hermitian ($\mathbf{G}_{\nu\nu} = \mathbf{G}_{\nu\nu}^\dagger$):

$$\mathbf{G}_{\nu\nu} = \frac{1}{M^2} \begin{pmatrix} g_{ee} & g_{e\mu} & g_{e\tau} \\ g_{e\mu}^* & g_{\mu\mu} & g_{\mu\tau} \\ g_{e\tau}^* & g_{\mu\tau}^* & g_{\tau\tau} \end{pmatrix}, \tag{2}$$

where M is the mass scale of the generic light mediator. Similarly, for the LRI sector, we define a separate Hermitian coupling matrix $\mathbf{G}_X$ governing the interactions with ordinary matter species X. These fundamental couplings dynamically generate the macroscopic effective potential $\mathbf{V}_{\rm LRI}$ as a function of the mediator mass and the cosmological density of the sourcing particles.

2 The cosmological quantum kinetic equation

Astrophysical neutrinos propagate over cosmological baselines in an expanding universe. We describe the neutrino ensemble using the 3×3 comoving differential number density matrix $\rho(E_0, z)$, where E_0 is the *observed* energy at Earth ($z = 0$). The physical energy at redshift z is $E_z = E_0(1 + z)$.

Transforming the Liouville-von Neumann equation to comoving coordinates using the FLRW relation $dt = -dz/\left[H(z)(1+z)\right]$, where $H(z) = H_0\sqrt{\Omega_m(1+z)^3+\Omega_\Lambda}$, yields the master transport equation

$$\frac{d\rho(E_0,z)}{dz} = \frac{i}{H(z)(1+z)}\left[\mathbf{H}_{\rm tot}(E_0,z),\rho(E_0,z)\right] - \frac{1}{H(z)(1+z)}\mathcal{C}[\rho(E_0,z)]. \tag{3}$$

Note on antineutrinos: The framework is identical for the antineutrino density matrix $\bar{\rho}(E_0,z)$, subject to the standard transformations $\mathbf{U}_{\rm PMNS} \to \mathbf{U}^*_{\rm PMNS}$ and $\mathbf{V}_{\rm LRI} \to -\mathbf{V}^*_{\rm LRI}$. This specific sign flip for the matter potential explicitly assumes a vector mediator. If the interaction were mediated by a scalar boson, the potential would not change sign between neutrinos and antineutrinos, whereas an axial-vector mediator would introduce different spin-dependent signs.

3 The coherent term: standard and non-standard potentials

The effective Hamiltonian $\mathbf{H}_{\rm tot}$ drives the coherent forward scattering. Because the physical momentum redshifts, the vacuum oscillation frequency must be evaluated dynamically at E_z:

$$\mathbf{H}_{\rm tot}(E_0,z) = \frac{1}{2E_0(1+z)}\mathbf{U}_{\rm PMNS}\mathbf{M}^2\mathbf{U}^\dagger_{\rm PMNS} + \mathbf{V}_{\rm SM}(z) + \mathbf{V}_{\rm LRI}(z). \tag{4}$$

Here, $\mathbf{U}_{\rm PMNS}$ is the Pontecorvo-Maki-Nakagawa-Sakata mixing matrix, and $\mathbf{M}^2 = {\rm diag}(0,\Delta m^2_{21},\Delta m^2_{31})$.

Standard Model boundary condition: The Standard Model matter potential is governed by the charged-current interactions with free electrons,

$$\mathbf{V}_{\rm SM}(z) = \sqrt{2}G_F n_{e,0}(1+z)^3{\rm diag}(1,0,0). \tag{5}$$

Crucially, because $\mathbf{V}_{\rm SM}(z)$ scales purely with local densities, its operational significance is dominant primarily for short-range constraints. Over macroscopic ultra-long-range cosmological baselines, the horizon-wide forward-scattering volume ensures that $\mathbf{V}_{\rm LRI}(z)$ dictates the dominant propagation eigenstates.

For ultra-light mediators, integrating the Yukawa potential over a continuous, homogeneous cosmological background yields an effective potential directly proportional to the physical number density $n_X(z) = n_{X,0}(1+z)^3$. This relies on the well-motivated assumption that the mediator is sufficiently light (e.g., $m_{Z_D} \lesssim 10^{-32}$ eV) such that its interaction range is comparable to the Hubble horizon, effectively smoothing out local coherent volumes. Furthermore, we assume a zero or negligible cosmic lepton asymmetry; otherwise, the CνB itself would source a non-vanishing coherent potential ($\mathbf{V}_{\nu\nu} \neq 0$).

4 The incoherent term: thermal resonance broadening

The collision operator $\mathcal{C}[\rho]$ dictates the attenuation and cascading of the high-energy flux due to scattering off the isotropic CνB, yielding

$$\mathcal{C}[\rho(E_0,z)] = -\frac{1}{2}\left\{\mathbf{\Gamma}_{\rm scat}(E_0,z),\rho(E_0,z)\right\} + \mathcal{P}_{\rm diag}\left(\int_{E_0}^{\infty} dE_0'\,\mathbf{W}(E_0'\to E_0,z)\rho_{\rm diag}(E_0',z)\right), \tag{6}$$

where $\mathcal{P}_{\rm diag}$ enforces absolute quantum decoherence during down-scattering (i.e., operating mathematically as $\mathcal{P}_{\rm diag}(\rho)_{ij} = \rho_{ij}\delta_{ij}$). This collision operator formulation assumes sudden, incoherent hard scattering, where the interaction timescale is much shorter than the neutrino oscillation length. This ensures the validity of separating the effective field theory into distinct coherent forward scattering and incoherent non-forward scattering regimes without macroscopic quantum interference between the two.

Identical particle phase space factor: To preserve exact cross-section normalization when integrated over an isotropic Fermi-Dirac distribution $f_{\rm FD}(p, T_\nu(z))$, a symmetry factor of $1/2$ must be included to account for identical initial-state background states.

The scattering rate matrix $\mathbf{\Gamma}_{\rm scat}$, governing flux attenuation, is determined by integrating the total cross-section,

$$(\mathbf{\Gamma}_{\rm scat})_{\alpha\beta}(E_0, z) = \sum_\gamma \int \frac{d^3p}{(2\pi)^3} f_{\rm FD}(p, T_\nu(z))\, \sigma(s(p, E_0, z), (\mathbf{G}_{\nu\nu})_{\alpha\gamma})\, (1 - \cos\theta). \tag{7}$$

Similarly, the transition matrix $\mathbf{W}(E_0' \to E_0, z)$, governing energy cascading, is defined by integrating the differential cross-section,

$$(\mathbf{W}(E_0' \to E_0, z))_{\alpha\beta} = \sum_\gamma \int \frac{d^3p}{(2\pi)^3} f_{\rm FD}(p, T_\nu(z))\, \frac{d\sigma(s', E_0, (\mathbf{G}_{\nu\nu})_{\alpha\gamma})}{dE_0}\, (1 - \cos\theta). \tag{8}$$

Here, θ is defined as the scattering angle between the incoming high-energy neutrino and the thermal background momentum in the cosmic rest frame. Note that the differential cross section $\frac{d\sigma}{dE_0}$ must be evaluated with respect to the observed energy, picking up a kinematic Jacobian factor of $(1 + z)$ relative to the physical frame.

The center-of-mass energy squared is defined continuously across the thermal phase space as $s(p, E_0, z) \approx 2pE_0(1 + z)(1 - \cos\theta)$ for the attenuation term, and $s' \approx 2pE_0'(1 + z)(1 - \cos\theta)$ for the initial state of the cascading term. This explicitly couples the energy redshifting to the thermal background, ensuring the realistic physical broadening of resonance dips. Kinematic limits within the differential cross-section naturally cause $\mathbf{W}$ to vanish outside the allowed two-body phase space.

5 Application: the dark-neutrino portal

In the $U(1)_D$ dark-neutrino portal model, active neutrinos interact with the dark sector exclusively through active-sterile mixing, parameterized by $U_{\alpha 4}$. Kinetic mixing vanishes in the electrically neutral intergalactic medium, meaning the long-range potential is sourced exclusively by cosmological neutrons.

The general 3×3 LRI interaction matrix is fully populated to accommodate arbitrary and independent values for all three sterile mixing components ($u_e \equiv |U_{e4}|$, $u_\mu \equiv |U_{\mu 4}|$, $u_\tau \equiv |U_{\tau 4}|$). This generalization preserves complex CP-violating phases, such that the matrix elements take the form $U_{{\rm LRI},\alpha\beta} \propto U_{\alpha 4} U_{\beta 4}^*$.

Furthermore, when propagating through the thermal cosmic neutrino background, the base mediator mass squared ($m_{Z_D}^2$) in the propagator must be augmented by the thermal Debye screening mass squared ($m_D^2(z)$) (see Section 7 for the explicit derivation). The modified macroscopic LRI potential matrix $\mathbf{V}_{\rm LRI}^{(n)}$ for neutrinos is thus defined as:

$$\mathbf{V}_{\rm LRI}^{(n)}(z) = +\frac{g_D \epsilon'}{m_{Z_D}^2 + m_D^2(z)} \frac{4 c_W m_Z^2}{g} \left(\frac{\sqrt{2}}{2} G_F n_{n,0} (1 + z)^3 \right) \begin{pmatrix} |U_{e4}|^2 & U_{e4} U_{\mu 4}^* & U_{e4} U_{\tau 4}^* \\ U_{\mu 4} U_{e4}^* & |U_{\mu 4}|^2 & U_{\mu 4} U_{\tau 4}^* \\ U_{\tau 4} U_{e4}^* & U_{\tau 4} U_{\mu 4}^* & |U_{\tau 4}|^2 \end{pmatrix}. \tag{9}$$

For antineutrinos, the potential $\bar{\mathbf{V}}_{\rm LRI}^{(n)}(z)$ undergoes a sign reversal alongside the complex conjugation of the active-sterile mixing elements:

$$\bar{\mathbf{V}}_{\rm LRI}^{(n)}(z) = -\frac{g_D \epsilon'}{m_{Z_D}^2 + m_D^2(z)} \frac{4 c_W m_Z^2}{g} \left(\frac{\sqrt{2}}{2} G_F n_{n,0} (1 + z)^3 \right) \begin{pmatrix} |U_{e4}|^2 & U_{e4}^* U_{\mu 4} & U_{e4}^* U_{\tau 4} \\ U_{\mu 4}^* U_{e4} & |U_{\mu 4}|^2 & U_{\mu 4}^* U_{\tau 4} \\ U_{\tau 4}^* U_{e4} & U_{\tau 4}^* U_{\mu 4} & |U_{\tau 4}|^2 \end{pmatrix}. \tag{10}$$

Dynamical Non-Standard Self-Interactions (NSSI):
To connect the generic incoherent collision operator $\mathcal{C}[\rho]$ defined in Section 4 to the specific $U(1)_D$ model, we map the generalized cross-section integral σ to a dynamical Breit-Wigner resonance formulation. Rather than approximating the Non-Standard Self-Interaction (NSSI) strictly as a static effective coupling matrix, the integration framework evaluates the scattering of incident high-energy neutrinos off the cosmic neutrino background via the resonance width:

$$\Gamma_{\text{width}} = \frac{\Sigma G \cdot m_{Z_D}}{12\pi}, \tag{11}$$

where ΣG encompasses the total effective coupling (accounting for both pure dark interactions and Standard Model interference) evaluated for the given flavor transitions:

$$(\mathbf{\Sigma G})_{\alpha\beta} = g_D|U_{\alpha 4}||U_{\beta 4}| \left(g_D|U_{\alpha 4}||U_{\beta 4}| + \epsilon' \frac{g}{c_W} \right). \tag{12}$$

This dynamical approach directly tracks the cosmological redshift evolution ($(1+z)^3$) of the background neutron density and integrates the exact momentum sequences of the thermal background (p, E_{bg}), ensuring the accurate capture of both energy cascading and resonant flux attenuation.

5.1 Benchmark textures (assuming vanishing CP phases)

While the general framework accommodates fully populated complex matrices, we can successfully recover the simplified 2×2 block-diagonal limits—matching the specific benchmark cases explored by Jana & Swain—by selectively setting specific mixing elements to zero and assuming vanishing CP phases. Let $u \equiv |U_{\alpha 4}| = |U_{\beta 4}|$ represent the common mixing magnitude for the active states. *These matrix reductions are strictly mathematically valid only if the active-sterile CP-violating phases vanish (i.e., $U_{\alpha 4}$ is purely real).*

The $e\mu$ sector benchmark ($|U_{e4}| = |U_{\mu 4}| \equiv u$, and $|U_{\tau 4}| = 0$):

$$\mathbf{V}_{\text{LRI}}^{(e\mu)} = \mathcal{A}_{\text{LRI}}(z) \begin{pmatrix} u^2 & u^2 & 0 \\ u^2 & u^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{\Sigma G}^{(e\mu)} = \mathcal{A}_{\text{NSSI}} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{13}$$

where the macroscopic potential amplitude $\mathcal{A}_{\text{LRI}}(z)$ (incorporating Debye screening and the neutrino/antineutrino sign) and the dynamical NSSI coupling amplitude $\mathcal{A}_{\text{NSSI}}$ are defined as:

$$\mathcal{A}_{\text{LRI}}(z) = \pm \frac{g_D \epsilon'}{m_{Z_D}^2 + m_D^2(z)} \frac{4 c_W m_Z^2}{g} \left(\frac{\sqrt{2}}{2} G_F n_{n,0} (1+z)^3 \right), \tag{14}$$

$$\mathcal{A}_{\text{NSSI}} = g_D u^2 \left(g_D u^2 + \epsilon' \frac{g}{c_W} \right). \tag{15}$$

The $e\tau$ sector benchmark ($|U_{e4}| = |U_{\tau 4}| \equiv u$, and $|U_{\mu 4}| = 0$):

$$\mathbf{V}_{\text{LRI}}^{(e\tau)} = \mathcal{A}_{\text{LRI}}(z) \begin{pmatrix} u^2 & 0 & u^2 \\ 0 & 0 & 0 \\ u^2 & 0 & u^2 \end{pmatrix}, \quad \mathbf{\Sigma G}^{(e\tau)} = \mathcal{A}_{\text{NSSI}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}. \tag{16}$$

The $\mu\tau$ sector benchmark ($|U_{\mu 4}| = |U_{\tau 4}| \equiv u$, and $|U_{e4}| = 0$):

$$\mathbf{V}_{\text{LRI}}^{(\mu\tau)} = \mathcal{A}_{\text{LRI}}(z) \begin{pmatrix} 0 & 0 & 0 \\ 0 & u^2 & u^2 \\ 0 & u^2 & u^2 \end{pmatrix}, \quad \mathbf{\Sigma G}^{(\mu\tau)} = \mathcal{A}_{\text{NSSI}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}. \tag{17}$$

6 Observables at Earth and parameter space

To confront the theoretical framework with high-energy astrophysical neutrino data from IceCube, we must compute the diffuse, continuous flux arriving at Earth. This requires simulating the cumulative emission from a cosmological population of sources distributed across redshift and incorporating the full multidimensional parameter space of both standard and new physics.

6.1 Astrophysical source modeling

We assume that the high-energy neutrinos are emitted by identical, steady-state astrophysical sources. The initial, unoscillated comoving density matrix injected at the source redshift z_s is defined by a power-law energy spectrum and an arbitrary initial flavor composition:

$$\rho(E_s, z_s) = \Phi_0 E_s^{-p} \mathrm{diag}(f_{e,S}, f_{\mu,S}, f_{\tau,S}), \tag{18}$$

where Φ_0 is a normalization constant, $E_s = E_0(1 + z_s)$ is the physical energy at emission, p is the free spectral index, and the initial flavor fractions satisfy $f_{e,S} + f_{\mu,S} + f_{\tau,S} = 1$. The parameters $f_{e,S}$ and $f_{\mu,S}$ are treated as freely varying to allow the model to remain agnostic to the specific hadronic or photohadronic production mechanisms at the source. This strictly diagonal structure correctly incorporates the fact that high-energy astrophysical neutrinos are produced isotropically and emerge unpolarized at the source.

The total diffuse density matrix at Earth, $\rho^{\oplus}_{\mathrm{diff}}(E_0)$, is computed by integrating the QKE solutions $\rho^{\oplus}(E_0, z_s)$ (evaluated at $z = 0$ for a source at redshift z_s) over the cosmological source distribution:

$$\rho^{\oplus}_{\mathrm{diff}}(E_0) = \int_0^{z_{\max}} \frac{dz_s}{H(z_s)} \frac{R(z_s)}{1 + z_s} \rho^{\oplus}(E_0, z_s). \tag{19}$$

The comoving source number density evolution, $R(z)$, is parameterized as proportional to the cosmic star-formation rate (SFR) modulated by an evolutionary power-law factor:

$$R(z) \propto \dot{\rho}_{\mathrm{SFR}}(z)(1 + z)^{\rho}, \tag{20}$$

where ρ is a free parameter governing the source density evolution relative to the standard SFR.

6.2 Standard neutrino parameters

The propagation relies heavily on the Standard Model mixing and mass structure. In our global framework, the absolute masses of the mass eigenstates (m_1, m_2, m_3) are treated as free parameters, bounded indirectly by oscillation data and the cosmological upper limit on the sum of neutrino masses $(\sum m_i)$.

Furthermore, the elements of the PMNS mixing matrix ($\mathbf{U}_{\mathrm{PMNS}}$) and the mass-squared splittings are allowed to float freely. To ensure realistic constraints, the active mixing angles $(\theta_{12}, \theta_{13}, \theta_{23})$, the Dirac CP-violating phase (δ_{CP}), and the splittings $(\Delta m^2_{21}, \Delta m^2_{31})$ are equipped with Gaussian priors pulled from the latest global oscillation fits (e.g., NuFIT).

6.3 Joint observable extraction

By evaluating the final diffuse density matrix at Earth, we can jointly probe new physics through two distinct observable channels.

First, the energy spectrum undergoes attenuation and cascading due to non-standard self-interactions. The total observable flux at a given energy is proportional to the trace of the density matrix,

$$\Phi_{\mathrm{tot}}(E_0) \propto \mathrm{Tr}[\rho^{\oplus}_{\mathrm{diff}}(E_0)], \tag{21}$$

which will exhibit absorption troughs at resonant energies dictated by the self-interactions and the background mass eigenvalues.

Table 1: Summary of free parameters in the generalized transport and dark-portal framework.

Parameter	**Description**	**Treatment / Prior**
	Astrophysical source parameters	
p	Source injection spectral index	Free
ρ	Source redshift evolution index (relative to SFR)	Free
$f_{e,S}, f_{\mu,S}$	Initial flavor fractions ($f_{\tau,S} = 1 - f_{e,S} - f_{\mu,S}$)	Free
	Standard neutrino parameters	
m_1, m_2, m_3	Absolute neutrino mass eigenstates	Free (cosmological bounds)
$\Delta m^2_{21}, \Delta m^2_{31}$	Mass-squared splittings	Free (NuFIT prior)
$\theta_{12}, \theta_{13}, \theta_{23}$	PMNS mixing angles	Free (NuFIT prior)
$\delta_{\rm CP}$	Dirac CP-violating phase	Free (NuFIT prior)
	New physics couplings (dark-portal benchmarks)	
u_e, u_μ, u_τ	Active-sterile mixing magnitudes ($\lvert U_{\alpha 4}\rvert$)	Free
m_{Z_D}	Dark vector mediator mass	Free
g_D	Dark gauge coupling	Free
ϵ'	$Z - Z_D$ mass mixing parameter	Free
	New physics couplings (most general generic case)	
$g_{ee}, g_{\mu\mu}, g_{\tau\tau}$	Diagonal NSSI effective couplings	Free (real)
$g_{e\mu}, g_{e\tau}, g_{\mu\tau}$	Off-diagonal NSSI effective couplings	Free (complex)
$g_{X,ee}, g_{X,\mu\mu}, g_{X,\tau\tau}$	Diagonal LRI effective couplings	Free (real)
$g_{X,e\mu}, g_{X,e\tau}, g_{X,\mu\tau}$	Off-diagonal LRI effective couplings	Free (complex)
M	Generic light mediator mass scale	Free

Second, the final flavor fractions at Earth are extracted from the normalized diagonal elements of the resulting density matrix:

$$f^{\oplus}_{\alpha}(E_0) = \frac{[\rho^{\oplus}_{\rm diff}(E_0)]_{\alpha\alpha}}{\mathrm{Tr}[\rho^{\oplus}_{\rm diff}(E_0)]}. \tag{22}$$

Scanning this comprehensive parameter space yields the predicted spectral distortions and energy-dependent flavor composition, allowing for a rigorous, simultaneous fit to IceCube's measured energy spectrum and flavor triangle allowed regions.

6.4 Summary of free parameters

To execute this multidimensional fit, the numerical solver must navigate a unified parameter space encompassing astrophysics, standard neutrino oscillations, and the dark-sector couplings. These parameters are summarized in Table 1.

7 Thermal screening by the Cosmic Neutrino Background

In finite-temperature field theory, a gauge boson propagating through a thermal plasma of charged fermions acquires an effective thermal mass, or Debye mass (m_D), due to vacuum polarization loop corrections. For the Z_D boson, the CνB acts as this thermal plasma. Because

the active neutrinos couple to the Z_D strictly through their mixing with the sterile state ν_4, they carry an effective dark charge $g_{\text{eff},\alpha} = g_D|U_{\alpha 4}|^2$.

If the Debye mass exceeds the vacuum mass of the mediator ($m_D > m_{Z_D}$), the infinite-range $1/r$ potential is exponentially suppressed into a short-range Yukawa potential $e^{-m_D r}/r$. Over cosmological baselines, this effectively screens the LRI sourced by distant matter, neutralizing its ability to induce coherent flavor transitions.

7.1 Calculation of the Debye mass and the modified potential

The Debye mass squared generated by a relativistic background of chiral fermions at temperature $T_{\nu_\gamma}(z)$ is given by $m_D^2 = \sum_\gamma g_{\text{eff},\gamma}^2 T_{\nu_\gamma}^2/6$. For the dark-neutrino portal, summing over the active flavors $\gamma \in \{e, \mu, \tau\}$ yields:

$$m_D^2(z) = \frac{g_D^2}{6} \sum_\gamma |U_{\gamma 4}|^4 T_{\nu_\gamma}^2(z), \tag{23}$$

where $T_{\nu_\gamma}(z) = T_{\nu_\gamma,0}(1+z)$ is the evolving temperature of the CνB.

As established in Section 5, this dynamically evolving thermal mass modifies the zero-momentum limit of the Z_D propagator, replacing $1/m_{Z_D}^2$ with $1/(m_{Z_D}^2 + m_D^2(z))$. This effectively screens the macroscopic potential $\mathbf{V}_{\text{LRI}}^{(n)}(z)$ whenever $m_D(z) \gg m_{Z_D}$.

7.2 The role of flavor asymmetries and potential textures

The summation in the Debye mass explicitly links the survival of the LRI to the alignment between the texture of the new physics and the flavor composition of the CνB.

Standard cosmology predicts a flavor-symmetric CνB ($T_{\nu_e} \approx T_{\nu_\mu} \approx T_{\nu_\tau}$). Under these conditions, the screening is largely texture-independent; as long as the Z_D couples to any active flavor, it encounters a thermal bath and acquires a mass.

However, if there exists a large cosmic lepton asymmetry (e.g., $n_{\nu_e} \gg n_{\nu_\mu,\nu_\tau}$), the background plasma is predominantly composed of ν_e. If the new physics is governed by a $\mu\tau$-sector benchmark texture ($|U_{e4}| = 0$), the Z_D boson is completely blind to the dominant ν_e background. In this orthogonal alignment, $m_D(z) \to 0$, the CνB fails to screen the interaction, and the long-range potential survives over gigaparsec distances. Conversely, an $e\mu$-textured LRI would suffer catastrophic screening from the exact same asymmetric background.

7.3 Phenomenological consequences: Mass regimes and observables

The inclusion of CνB thermal screening creates a phenomenological bifurcation for the observable signatures at IceCube, cleanly partitioning the parameter space into two disjoint regimes:

- **The ultra-light regime ($m_{Z_D} \lesssim 10^{-30}$ eV):** To generate macroscopic refractive flavor shifts at Earth via the coherent matter potential, the vacuum mediator mass must be tiny. However, even for tiny couplings ($g_{\text{eff}} \sim 10^{-10}$), the present-day Debye mass induced by the CνB is $m_D \sim 10^{-14}$ eV. Because $m_D \gg m_{Z_D}$, the LRI is screened. *Observable impact:* The coherent LRI potential $\mathbf{V}_{\text{LRI}}$ vanishes, and the flavor-rotation phase shifts are erased, returning the propagation to standard vacuum oscillations (unless rescued by an orthogonal texture/asymmetry alignment).

- **The MeV regime ($m_{Z_D} \sim \mathcal{O}(\text{MeV})$):** To place the s-channel resonant scattering dip inside the IceCube energy window (10 TeV – 10 PeV), the mediator mass must be in the MeV scale to satisfy the kinematic condition $s \approx m_{Z_D}^2$. Because $m_{Z_D} \sim 10^6$ eV $\gg m_D$, thermal screening from the CνB is negligible. *Observable impact:* The resonant

absorption dips in the energy spectrum remain intact. Furthermore, the localized, energy-dependent flavor transitions corresponding to these dips survive, as they are driven purely by dissipative differential absorption rather than coherent refractive phase shifts.